VOÏLA PLATFORM SPECIFICATION

Comprehensive Technical Reference: Complete Feature Catalog, Mathematical Metrics, ML/NLP Algorithms, Agentic AI Tooling & Relational Database Schemas

Platform Name:	**Voilà — Voice of Customer Signal Intelligence Platform**
Release Version:	v2.4.0 (Enterprise Production Build)
Active Scale:	**105,000 Ingested Customer Conversations**
Core Stack:	FastAPI, React 18, PostgreSQL 15, Qdrant Vector DB, Snowflake, AWS Bedrock
System Capabilities:	Real-Time SLA Triage, Root-Cause NLP, Conversational AI Copilot, S3 Data Lake
Author / Architect:	Voilà Engineering & AI Architecture Group

Executive Overview:
Voilà is an enterprise-grade intelligence platform that converts high-volume, unstructured customer support dialogues into real-time operational diagnostics, root-cause friction themes, and executive decision policies. The system continuously evaluates 105,000+ interactions across response velocity, First Contact Resolution (FCR), reopen rates, and sentiment distress.

1. Complete Platform Feature Catalog

1.1 Executive Dashboard & Operational Intelligence (/)

- **Live Telemetry Grid:** Instant KPI cards tracking Total Interactions (105,000), First Contact Resolution Rate (53.7%), Average Response Time (133.7m), Ticket Reopen Rate (44.5%), Negative Friction Share (24.2%), and CSAT Index (3.8/5.0).
- **SLA Latency Distribution Matrix:** Multi-bucket response latency analysis categorizing tickets into <15m, 15-60m, 1-4h, and >4h with dynamic percentage breakdowns.
- **Sentiment & Friction Donut:** Three-tier sentiment polarity breakdown measuring positive satisfaction vs negative friction.
- **Service Velocity Trend Analysis:** Continuous temporal response curve tracking velocity shifts across hourly, daily, and monthly intervals.
- **Topic Quadrant Matrix (2D Scatter):** 2-dimensional scatter visualization plotting *Friction Severity vs Ticket Volume* to separate high-impact critical blockers from low-severity high-volume noise.
- **Unified Regional Intelligence:** Geographic breakdown comparing SLA latency and sentiment across North America, Europe, Asia Pacific, Latin America, and Middle East & Africa.
- **Priority Action Board:** Autonomous prioritization engine generating P0, P1, and P2 operational interventions mapped to Engineering, Support Operations, and Billing teams.
- **Spike Detection Banner:** Statistical anomaly detector alerting teams when volume or negative tone spikes beyond 2.5 standard deviations.
- **Multi-Format Custom Export:** One-click generation and instant download of customized executive briefs in **PDF, CSV, JSON, and Markdown** formats.

1.2 Topic Clusters & NLP Deep-Dive (/topics)

- **Automated Unsupervised Clustering:** BERTopic and class-based TF-IDF pipeline grouping conversations into semantic friction clusters.
- **Pre-Configured Core Clusters:** Deep diagnostics for *Poor customer support, Application malfunction, Delivery & Order logistics, Billing & Invoices, Connectivity/Network failures, and Account Access/2FA*.
- **Verbatim Customer Quotes:** Anonymized, live dialogue evidence with timestamps, author roles (customer vs agent), and sentiment polarity.
- **Pain Score & Escalation Risk:** Mathematically weighted operational index quantifying severity based on volume, negative tone, and response delay.
- **Root-Cause Breakdown:** Systematic engineering and support root-cause analysis for each identified topic cluster.

1.3 Ask Data — Gemini / ChatGPT Conversational Copilot (/ask)

- **Conversational Chat Experience:** Streamlined chat layout with Voilà sparkle brand avatar, sleek user bubbles, and Gemini-style animated reasoning indicator.
- **Natural Persona Intelligence:** Contextual answers for identity queries ('what is your name'), capabilities ('what can you do'), politeness ('thank you'), greetings, and farewells.
- **Root-Cause & Volume Diagnosis:** Specialized reasoning engine capable of diagnosing specific numbers (e.g. 15,700 messages driving poor customer support friction).
- **Action Toolbar & Collapsible Telemetry:** Copy-to-clipboard button, feedback buttons, and a collapsible accordion containing visual KPI charts.
- **Query Audit History:** Persistent history sidebar with single-item deletion on hover and a global 'Clear All' action.

1.4 Streaming Ingestion & S3 Data Pipeline (/upload)

- **Large-Scale File Ingestion:** Drag-and-drop CSV and Excel uploader supporting millions of rows without browser memory exhaustion.
- **Dual Ingestion Architecture:** In-Memory RAM Engine for sub-second small dataset uploads + Chunked Streaming Engine (20,000 rows/chunk) with zero OOM risk.
- **AWS S3 Object Storage Data Lake:** Automated cloud data lake streaming with bucket storage and S3 key management.
- **Dataset Catalog & Run Management:** Instant switching between dataset runs and a permanent delete confirmation modal (DELETE /analytics/runs/{run_id}).

1.5 Dataset Delta & Multi-Period Comparison (/compare)

- **Side-by-Side Variance Analysis:** Compare two distinct uploaded runs or two calendar years (e.g. 2017 vs 2018).
- **Automated Delta Calculations:** Live delta indicators for Delta Resolution (%), Delta Response Time (mins), Delta Negative Sentiment, and Delta Volume.
- **Emergent Topic Detection:** Flags newly emerging complaint clusters and tracks resolved issues over time.

1.6 Global UX & Security Infrastructure

- **Hardware-Accelerated 60 FPS Sidebar:** Smooth GPU-composited collapsible navigation sidebar with official Voilà 'V' brand mark.
- **Persistent Floating AI Copilot:** Accessible from any screen via a floating launcher bubble in the bottom-right.
- **Global Filter Bar:** Slices telemetry across Time Period, Year, Month, Date Range, Brand/Company, Product, and Region.
- **JWT Security:** Secure authentication with token persistence, protected route wrappers, and user profile management.

2. Operational Metrics & Mathematical KPI Models

Every operational metric in Voilà is rigorously defined and computed from raw relational PostgreSQL conversation records:

1. First Contact Resolution Rate (FCR / Resolution Rate)	**Active Baseline: 53.7%** | **Target SLA: ≥ 75.0%**
Mathematical Formula:	$FCR\ (\%) = (N_{resolved} / N_{total}) \times 100$
Variable Definitions:	• $N_{resolved}$ = Count of conversations marked as resolved on first customer interaction. • N_{total} = Total count of inbound customer conversations.
Calculation Example:	In active dataset: 56,385 resolved / 105,000 total × 100 = **53.7%**.

2. Mean Response Latency (MRT)	**Active Baseline: 133.7 mins** | **Target SLA: ≤ 15.0 mins**
Mathematical Formula:	$MRT = (1 / N) \times \sum (T_{reply,\ i} - T_{inbound,\ i})$
Variable Definitions:	• $T_{reply,\ i}$ = Timestamp of first official support agent response. • $T_{inbound,\ i}$ = Timestamp of initial customer complaint message. • N = Total evaluated interaction pairs with valid response timestamps.
Calculation Example:	Sum of latency across 105,000 interactions = 14,038,500 minutes ÷ 105,000 = **133.7 minutes**.

3. Ticket Reopen Rate (CRR)	**Active Baseline: 44.5%** | **Target SLA: ≤ 20.0%**
Mathematical Formula:	$CRR\ (\%) = (N_{reopened} / N_{resolved}) \times 100$
Variable Definitions:	• $N_{reopened}$ = Count of tickets where customer sent follow-up messages after closure. • $N_{resolved}$ = Count of tickets previously marked resolved.
Operational Meaning:	Measures premature resolution friction. High reopen rate (44.5%) indicates agents close tickets before full technical fix is verified.

4. Negative Friction Share (NFS)	**Active Baseline: 24.2%** | **Target SLA: ≤ 10.0%**
Mathematical Formula:	$NFS\ (\%) = (N_{polarity < -0.10} / N_{total}) \times 100$
Variable Definitions:	• $N_{polarity < -0.10}$ = Count of interactions classified with negative sentiment polarity. • N_{total} = Total interaction volume in evaluated window.
Calculation Example:	25,410 negative interactions / 105,000 total = **24.2% Negative Friction Share**.

5. Composite CSAT Quality Index	**Active Baseline: 3.8 / 5.0** | **Target SLA: ≥ 4.5 / 5.0**
Mathematical Formula:	$CSAT = 1.0 + 4.0 \times [0.60 \times (FCR / 100) + 0.40 \times (1 - NFS / 100)]$
Weighting Rationale:	• 60% weight on operational resolution speed (FCR). • 40% weight on emotional customer tone and absence of negative friction (1 – NFS).
Calculation Example:	1.0 + 4.0 × [0.60 × 0.537 + 0.40 × (1 – 0.242)] = 1.0 + 4.0 × [0.3222 + 0.3032] = **3.8 / 5.0**.

6. Topic Pain Score (TPS)	**Benchmark Scale: 0 to 20,000 pts** | **Critical: > 5,000 pts**
Mathematical Formula:	$TPS = Volume \times (NegPercentage / 100) \times [1 + (AvgResponseTime / 100)]$

Operational Use:	Combines volume scale, customer distress, and SLA delay into a single priority score. Used to rank top complaint clusters for engineering sprint hotfixes.

3. Machine Learning & NLP Algorithms

3.1 Text Cleaning & Preprocessing Pipeline

Every raw customer interaction undergoes a 5-step normalization sequence:

1. **PII & Handle Scrubbing:** Strips user mentions (@TwitterHandle), URLs, tracking tokens, and ticket IDs.
2. **Case Normalization & Contraction Expansion:** Converts text to lowercase while expanding conversational contractions (e.g. *"can't"* → *"cannot"*).
3. **Emoji & Sentiment Token Retention:** Preserves emotional emoticons (e.g. ■, ■, ❤■) as high-value polarity tokens.
4. **Lemmatization & Stopword Filtering:** Strips generic English stopwords while preserving operational verbs (e.g. *"crash"*, *"refund"*, *"reopen"*).

3.2 Semantic Dense Embeddings & Vector Search

- **Embedding Model:** 384-dimensional dense vectors generated via sentence-transformers/all-MiniLM-L6-v2.
- **Cosine Similarity Metric:** Distance between query vector u and document vector v is computed as:
  $$\mathrm{CosineSimilarity}(u, v) = (u \cdot v) / (||u|| \times ||v||)$$
- **Vector Database Indexing:** Indexed in **Qdrant** using HNSW (Hierarchical Navigable Small World) graphs with sub-millisecond retrieval latency.

3.3 Class-Based TF-IDF (c-TF-IDF) Keyword Extraction

To extract descriptive cluster keywords across grouped topics, the platform uses class-based TF-IDF:

$$W_{t,c} = \mathrm{tf}_{t,c} \times \log [1 + (A / f_t)]$$

where $\mathrm{tf}_{t,c}$ is the frequency of word t in cluster c, A is the average number of words per cluster, and f_t is the total frequency of word t across all clusters.

3.4 Sentiment Polarity Classification Algorithm

- **Hybrid Polarity Engine:** Combines lexicon-based VADER rules with transformer self-attention scoring.
- **Continuous Score:** Maps tone to continuous polarity $s \in [-1.0, +1.0]$.
- **Threshold Boundaries:**
  - **Positive Tone:** $s \geq +0.10$
  - **Neutral Tone:** $-0.10 < s < +0.10$
  - **Negative Friction:** $s \leq -0.10$
  - **Critical Customer Distress:** $s < -0.60$ (Triggers P0 Escalation Rule)

3.5 Statistical Spike & Anomaly Detection (Z-Score Modeling)

The platform detects abnormal volume surges or negative sentiment spikes using rolling Z-Score variance:

$$Z = (X_t - \mu_{24h}) / \sigma_{24h}$$

- μ_{24h} = 24-hour moving average of conversation volume or negative sentiment.
- σ_{24h} = 24-hour moving standard deviation.
- **Alert Trigger:** When $Z > 2.50$ (99% statistical confidence interval), Voilà generates an automated **Spike Alert Banner** on the dashboard.

4. Agentic AI Reasoning Architecture & Tools

Voilà Copilot is powered by an autonomous, multi-tool agentic loop designed for enterprise reliability, speed, and strict factual grounding:

Lifecycle Phase	Responsible Engine	Detailed Execution Flow
1. Input Validation	QueryValidator	Inspects input question for data sufficiency, explicit dataset fields, time periods, and intent categorization (greeting, persona, KPI, root-cause, policy).
2. Tool Planning	DecisionEngine	Determines required toolchain (PostgreSQL Fast Engine, Qdrant Vector DB, Snowflake Data Warehouse, Analytics Hub).
3. Parallel Execution	ToolOrchestrator	Executes parallel sub-millisecond database queries, full-text lexical matches, and vector similarity retrieval.
4. Context Grounding	ContextBuilder	Aggregates verified database KPIs, cluster breakdowns, and verbatim customer quotes into an executive grounded prompt.
5. LLM Synthesis	BedrockClient	Generates concise, natural language response via AWS Bedrock Mantle / Instant local grounded fallback (0.6s circuit breaker).
6. Output Validation	ResultValidator	Verifies data confidence (MEASURED vs NO_DATA_AVAILABLE), checks for hallucinations, and returns validated AgentResponse.

4.1 Dedicated Tool Ecosystem

- **PostgreSQL Fast Analytics Tool:** Sub-5ms indexed queries across 105,000 customer records, computing dynamic aggregations, SLA distributions, and cluster stats with zero cold starts.
- **Qdrant Vector DB Tool:** 384-dimensional cosine similarity semantic embedding tool with non-blocking socket checks (<0.15s) for instant customer dialogue retrieval.
- **Snowflake Data Warehouse Tool:** Cloud warehouse integration with offline short-circuit safeguards preventing blocking timeouts.
- **AWS Bedrock LLM Client:** Grounded reasoning with 0.6s circuit-breaker and instant local executive synthesis.

4.3 The 7-Pillar RAG & Agent Query Intelligence Framework

Voilà implements an end-to-end 7-pillar query intelligence engine that eliminates classic RAG edge-case failures without relying on fragile keyword matchers:

Pillar / Problem	Algorithmic Solution	Mathematical Formulation & Pipeline Flow
Pillar 1: Typos	Domain-Aware Spell Normalizer	Levenshtein distance ≤ 2 matching against English & support vocabulary with repeated character collapse (*'phne frezing'* → *'phone freezing'*).
Pillar 2: Out-of-Domain	Post-Retrieval Relevance Validation	Evaluates evidence grounding post-retrieval; triggers **Topic Focus Policy Steer-Back** with proactive prompt chips if unrelated.
Pillar 3: Gibberish	Pre-Embedding Entropy & Pattern Filter	Shannon character entropy $H(X)$ & consonant-vowel ratio filters catch random keystrokes (*'xyz123 qwerty'*) before generating embeddings.

Pillar 4: Generic Queries	Query Specificity Check	Detects under-specified single-word prompts (*'phone'*, *'app'*) and prompts the user with interactive, domain-specific clarification chips.
Pillar 5: Multi-Intent	Query Decomposition & RRF Fusion	Splits compound queries → parallel retrieval → **Reciprocal Rank Fusion**: $\textit{RRF\_Score}(d) = \sum [1 / (60 + \textit{Rank}_q(d))]$
Pillar 6: Negation	Focal Extraction & Exclusion Filter	Extracts positive target topic for vector encoding while applying a negative exclusion filter to prevent semantic inversion.
Pillar 7: Multi-Signal	Composite Relevance Confidence	Avoids rigid threshold cliffs by blending signals: $S_{\text{composite}} = 0.50 \cdot S_{\text{vec}} + 0.30 \cdot S_{\text{lex}} + 0.20 \cdot S_{\text{topic}}$

5. Relational Database Architecture (PostgreSQL)

The platform stores all raw dialogues, normalized metrics, and query logs across 8 high-performance relational tables:

Table Name	Key Columns & Types	Purpose & Telemetry Stored
dataset_runs	run_id (VARCHAR PK) user_id (VARCHAR) uploaded_at (TIMESTAMP) total_records (INT) source_name (VARCHAR) status (VARCHAR)	Catalog of uploaded dataset runs, total row counts, storage paths, and processing status.
customer_conversations	id (INT PK) run_id (VARCHAR FK) tweet_id (VARCHAR) author_id (VARCHAR) inbound (BOOLEAN) timestamp (TIMESTAMP) text (TEXT) response_time_minutes (FLOAT)	105,000+ raw and clean customer dialogues, author roles (customer vs agent), and response times.
kpi_runs	id (INT PK) run_id (VARCHAR FK) total_conversations (INT) resolution_rate (FLOAT) avg_response_time (FLOAT) reopen_rate (FLOAT) csat_score (FLOAT)	Pre-aggregated executive metrics for fast dashboard rendering with zero runtime compute.
kpi_topics	id (INT PK) run_id (VARCHAR FK) cluster_name (VARCHAR) topic_keywords (TEXT) volume (INT) negative_complaints (INT) pain_score (FLOAT)	NLP friction clusters, volume, sentiment share, pain scores, and escalation case counts.
kpi_sentiment	id (INT PK) run_id (VARCHAR FK) sentiment (VARCHAR) count (INT) percentage (FLOAT)	Distribution counts and percentages for positive, neutral, and negative tones.
agent_conversations	id (INT PK) timestamp (TIMESTAMP) user_id (VARCHAR) question (TEXT) query_type (VARCHAR) answer (TEXT) status (VARCHAR)	Conversational Copilot query audit logs, query types, and generated AI responses.
agent_tools	id (INT PK) agent_conversation_id (INT FK) tool_name (VARCHAR)	Audit trail mapping which specific tools (Postgres, Qdrant, Snowflake, Bedrock) executed for each query.